# Simulated annealing (SA)

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Need Picture

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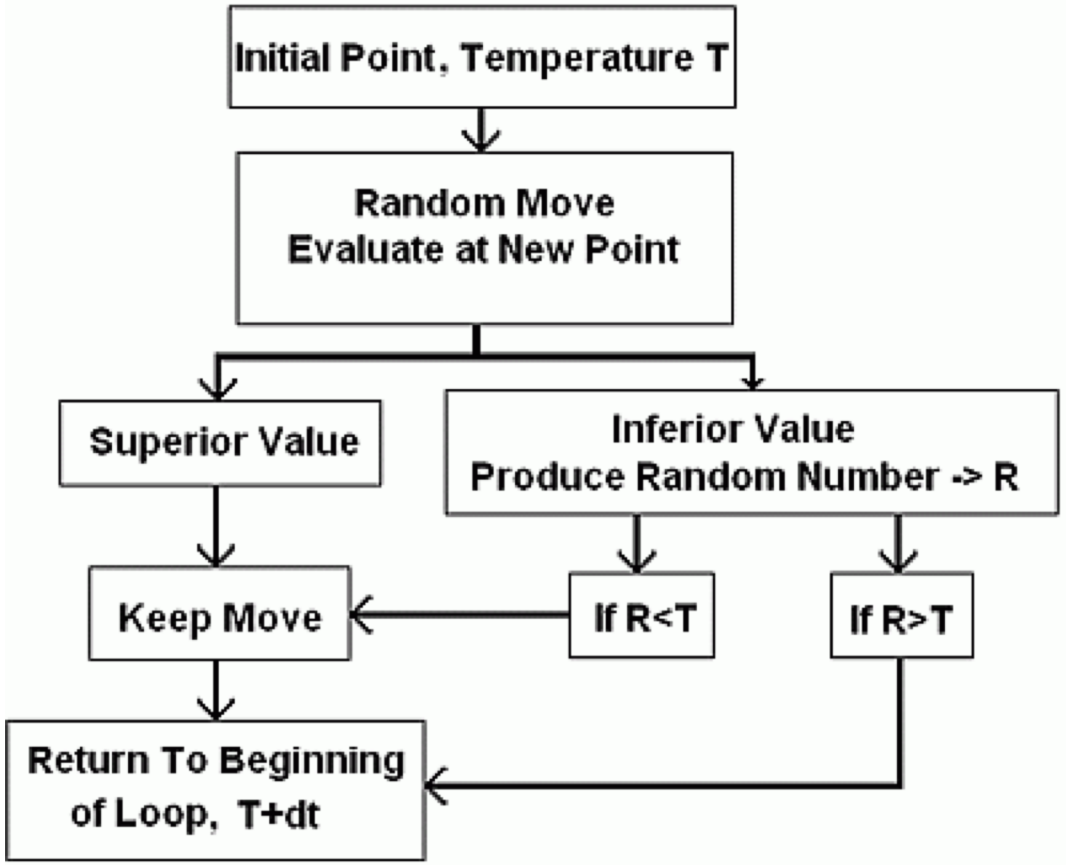
Annealing (metallurgy) - heating and controlled cooling of a material to increase the size of its crystals and reduce their defects.

**Simulated Annealing (SA)** - a probabilistic technique for approximating the global optimum of a given function in a large search space.

Slow cooling implemented in the Simulated Annealing algorithm is interpreted as a **slow decrease in the probability of accepting worse solutions** as the solution space is explored.

Note: accepting worse solutions is a fundamental property of metaheuristics because it allows for a more extensive search for the optimal solution.

The method is an adaptation of the Metropolis–Hastings algorithm, a Monte Carlo method to generate sample states of a thermodynamic system, invented by **M.N. Rosenbluth** and published by **N. Metropolis** et al. in 1953.